

Forecasting crude oil futures prices using BiLSTM-Attention-CNN model with Wavelet transform

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ABSTRACT

In this study, a novel hybrid model for forecasting crude oil futures price time series is proposed. The combination of Bidirectional long short-term memory network (BiLSTM), Attention mechanism and Convolution neural network (CNN) is used. Based on the principle of deep learning (DL) and wavelet transform (WT), the time series of crude oil futures price are decomposed and reconstructed into a low-frequency main sequence and several high-frequency noise sequence. Then, the subsequences obtained by decomposition are predicted by BiLSTM-Attention-CNN model in turn. Finally, with several evaluating indicators (RMSE, MAPE, MAE and R^2), the forecasting errors of proposed model and other comparative models are evaluated and compared. The experimental results based on two types of crude oil futures showed the novel model outperforms other related comparative models under different training sets lengths and the modification of Diebold–Mariano test (MDM) proved that the loss functions of constructed model are statistically significant in this paper. In general, the innovative combination forecasting model proposed in this paper is a promising technology for government agencies, investors and related enterprises.

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1. Introduction

Common crude oil products, such as gasoline and natural gas, are usually dealt in the financial market as futures contracts. Crude oil futures is the largest commodity futures, which plays the essential role in the financial market [1] and has a huge impact on the future planning, risk management and expenditure scale of enterprises. In addition, due to the importance of the crude oil futures market and its close relationship with some other financial markets (such as precious metal, stocks and exchange rates), there is a large amount of academic literatures on it [2–4]. The fluctuation of crude oil futures price will affect all aspects of our life and even hinder the development of national economy. Therefore, accurate and reliable forecasting of crude oil futures price is a very important issue. During the recent year, many scholars are looking for more accurate prediction models to forecast it, which is also the purpose of this paper.

There are some traditional methods to predict the price of crude oil futures, such as the autoregressive integral moving average (ARIMA) [5,6], random walk (RW) [7], generalized autoregressive conditional heteroscedasticity (GARCH) [8,9] and vectorial

autoregression (VAR) [10]. These traditional methods have good prediction effect for linear correlation variables, but they cannot catch the nonlinear characteristics of financial time series in the real world. Therefore, if we continue to use these methods, the prediction effect may not be further improved [11].

Because of the limitations of traditional models, a number of nonlinear artificial intelligence and deep learning methods emerge as the times require, and can be applied to time series prediction tasks. Such as artificial neural network (ANN) [12], support vector machine (SVM) [13] and back propagation neural network (BPNN) [14], these methods and models can solve the nonlinear problem of crude oil futures price time series, because many optimization algorithms are embedded [15–18]. Since price forecasting in the financial market is not only relevant to the series of the present time, but also to the previous data, if only the most recent point in time data is used, the information carried by the early data will be lost. Therefore, recurrent neural network (RNN) [19,20] is proposed. Unlike the traditional ANN, some connections between hidden units are made by RNN, which allows the neural structure to retain memories of recent states, so it is well suited for the tasks of time series prediction [21,22]. However, since the problems of gradient explosion and gradient disappearance in RNN, an improved Long Short-Term Memory network (LSTM) [23,24] based on RNN has been proposed. The information can be screened out through the various gate structures

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of LSTM, and useful information can be gleaned from historical data during the training process.

Compared with LSTM, Bidirectional LSTM (BiLSTM) could make use of reverse and forward information, which is made up of two layers of LSTM [25]. Therefore, it is especially suitable for solving time series forecasting problems. For example, Sima et al. compared LSTM with BiLSTM and predicted the financial time series. The result showed that the BiLSTM provides more accurate prediction effect than ARIMA and LSTM models [26]. Kulshrestha et al. combined BiLSTM with Bayesian Optimization (BO) to forecast the number of tourists (demand) to Singapore every quarter, which proves the effectiveness of BiLSTM [27].

Moreover, on account of the different contributions of historical data to data points at different times, and the current neural network models (including BiLSTM) have the same contribution to each time point to be predicted, so the attention mechanism is proposed [28]. To further improve the prediction accuracy of the model, Attention mechanism is usually embedded in deep learning model, and it can obtain effective information and significant spatiotemporal features from the new coding sequence. Zhou et al. used regularized self-attention (RSA) mechanism to extract temporal and spatial features from precious metal price, thus enhancing the learning process of precious metal price data [29]. In addition, one of the flaws of the traditional neural network is the poor scalability caused by the complete connectivity of neurons, which is overcome by the convolution neural network (CNN). CNN is a commonly applied deep learning model in the domain of image recognition and it is a feedforward neural network proposed by Lecun et al. [30]. Compared with the traditional neural network, CNN improves the efficiency of the algorithm and reduces the number of parameters. The architecture of typical CNN consists of a convolution layer, a pool layer and a full connected layer [31]. As many literatures have proved, CNN has the advantages of extraction and recombination [32,33]. Li et al. conducted experiments on the historical data of cogeneration enterprises, designed and adjusted the model structure, and finally built an effective time series prediction model (CNN-LSTM based on attention mechanism). The experimental prediction results were better than ARIMA, SVR and single neural network model [34]. According to the literature review above, this paper proposes a BiLSTM-Attention-CNN deep learning combination model to predict the daily closing price of crude oil futures. In this model, BiLSTM inherits the advantages of LSTM, which can better handle with the long-term dependence of time series. Attention mechanism layer can extract effective information and pay attention to the trend and change of time series, while CNN layer can capture the hierarchical structure of data.

Besides, high noise, serious nonlinearity and nonstationarity determine the complexity and difficulty of crude oil futures forecasting. With the development of wavelet analysis theory, wavelet transform (WT) is extensively applied in signal analysis and processing because of its multi-scale analysis characteristics and the superiority to characterize the local characteristics of signals in time–frequency domain [35,36]. Generally speaking, it decomposes a signal into several subsequences to make it more predictable. In order to overcome the shortcoming that Fourier transform cannot analyze non-stationary signals, Mallat proposed a wavelet decomposition method Mallat algorithm in 1989, which decomposes and reconstructs the signal [37]. It is similar to short-time Fourier transform (STFT). But the difference between STFT and WT is that STFT divides the signal into windows, and FFT (fast Fourier transform) is applied to each segment, while wavelet transform directly changes the basis of Fourier transform. In other words, the basis of infinite trigonometric function is replaced by the attenuated wavelet basis of finite length. In the past few years, wavelet analysis has also been introduced into the

economic and financial fields, and shows its unique advantages [38,39].

Wavelet transform is often combined with statistical model and neural network. Mishra et al. proposed a new solar energy short-term prediction model based on the learning ability of deep learning (DL) technology of LSTM and the concept of wavelet transform (WT), which proved the effectiveness of the proposed WT-LSTM model [40]. Alipoura et al. proposed an innovative method based on the integration of deep neural network and wavelet transform. Compared with the benchmark model, the result shows that the model has higher prediction accuracy [41]. In order to further improve the accuracy of wind speed prediction, He et al. proposed two new hybrid models which combined signal processing and deep learning [42]. Firstly, wavelet transform is applied to decompose and reconstruct the wind speed series. Then, Deep belief network (DBN) is applied to capture high-dimensional features.

In this study, we propose a BiLSTM-Attention-CNN deep learning combination model with wavelet transform to predict the daily closing price of crude oil futures. Firstly, two kinds of crude oil futures (WTI and Brent) daily closing price time series are decomposed and reconstructed into a low-frequency main signal with a number of high-frequency noise signals by wavelet transform. Then, all the processed subsequence data are input into the BiLSTM firstly. Next, the new encoding sequence will be sent to the attention mechanism layer to enhance temporal features. Finally, the CNN is employed to extract the spatial features from the output data to generate the final output. The paper uses MAPE, MAE, RMSE and R^2 loss functions to estimate prediction performance of the models, and the WT-BiLSTM-Attention-CNN model is further verified superiority via MDM test. Mainly, the purposes of this paper are to answer two questions through empirical experiment: (1) Whether WT-BiLSTM-Attention-CNN can reliably and precisely predict the crude oil futures price or not? (2) Supposing that the last question could be testified, whether the WT-BiLSTM-Attention-CNN shows the best results in term of both WTI and Brent crude oil futures prices?

The arrangement of the rest of this paper is as follows. Section 2 describes the combination model and decomposition method used to predict the closing price of crude oil futures. Section 3 is applied to WTI and Brent crude oil futures daily closing price time series by the model proposed, and compared with other models. Finally, Section 4 summarizes the whole work and puts forward some policy suggestions.

2. Methodology

2.1. Wavelet transform

WT as a kind of non-stationary signal processing technology, we can get the signals of different frequencies through time–frequency conversion. The formula of continuous wavelet transform (CWT) is as follows:

$$CWT(a, \tau) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \cdot \psi\left(\frac{t-\tau}{a}\right) dt \quad (1)$$

where $f(t)$ is the original signal, $\psi(t)$ is a mother wavelet, a is the scale, which controls the expansion and contraction of the wavelet function, and τ is the translation quantity to control the translation of the wavelet function. The steps of wavelet transform are as follows:

- (1) Select a wavelet function $\psi(t)$ and scale a .
- (2) Calculate the wavelet coefficient from the starting position of the original signal using the wavelet transform formula (that is, calculate the similarity C , C represents the similarity between the wavelet function and a certain signal).

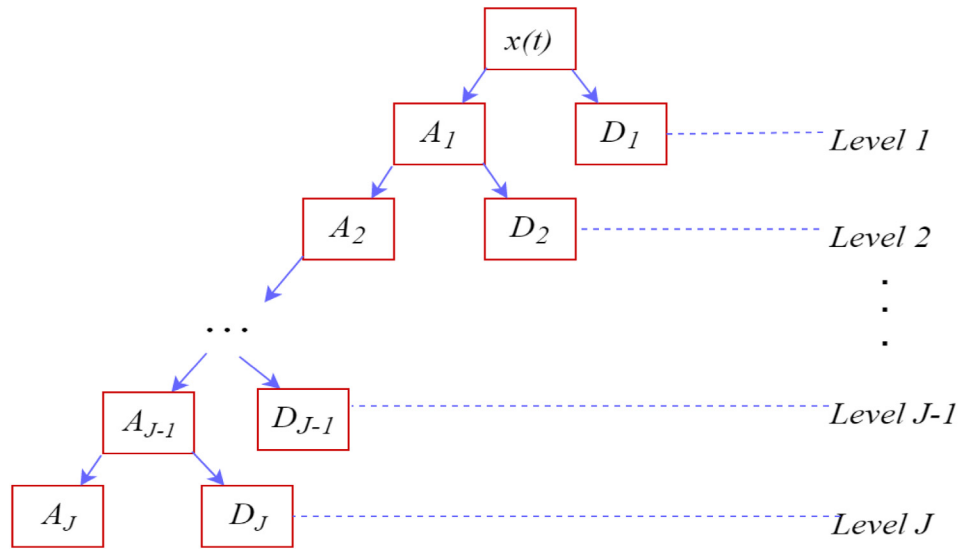


Fig. 1. Schematic diagram of wavelet decomposition and reconstruction.

(3) Continuously change the parameter τ , shift the wavelet to the right, and repeat the first and second steps until the whole signal is covered (i.e., scan and calculate along the time axis).

(4) Change the value of scale a to scale the wavelet, and repeat steps (2) and (3) (i.e. complete the scanning calculation of the frequency axis).

(5) Repeat the above four steps for all scale scaling (complete all scanning in time domain and frequency domain).

Any continuous wavelet transform will produce extremely detailed information, and soon become difficult to calculate. Therefore, discrete wavelet transform (DWT) is used in practice [37]. The formula of discrete wavelet transform is as follows:

$$DWT(m, n) = 2^{-m/2} \sum_{t=0}^{T-1} x(t) \cdot \Psi\left(\frac{t - n \cdot 2^m}{2^m}\right) \quad (2)$$

where $x(t)$ is the original signal, $\Psi(t)$ is a mother wavelet, m and n are similar to a and τ in Eq. (1), respectively.

DWT is to map a given time series $x(t)$ to its wavelet coefficients, and then inversely reconstruct the wavelet coefficients to form two types of components, one is the approximate component of low frequency and the other is the detail component of high frequency. As shown in the following formula, the initial signal can be represented as the sum of signals of different frequencies.

$$x(t) = \sum_{k=1}^T c_{j,k} \Psi_{j,k}(t) + \sum_{j=1}^J \sum_{k=1}^T d_{j,k} \Psi_{j,k}(t) \quad (3)$$

where $c_{j,k}$ and $d_{j,k}$ represent the approximate coefficient and detail coefficient when the scale is j and the location is k , respectively. T is the length of the time series, J is the number of levels of decomposition.

For the time series $x(t)$, it can be decomposed into the sum of high frequency component $D_1(t)$ and low frequency component $A_1(t)$, that is, $x(t) = A_1(t) + D_1(t)$. As shown in the Fig. 1, in order to get a finer scale decomposition, $A_1(t)$ is treated in the same way: $A_1(t) = A_2(t) + D_2(t)$. Therefore, Eq. (3) can be simplified to

$$x(t) = A_J(t) + \sum_{j=1}^J D_j(t) \quad (4)$$

where $A_j(t) = \sum_{k=1}^T c_{j,k} \Psi_{j,k}(t)$ and $D_j(t) = \sum_{k=1}^T d_{j,k} \Psi_{j,k}(t)$. In other words, after DWT, $x(t)$ can be expressed by the sum

of a low frequency approximate signal $A_j(t)$ and several high frequency detail signals $D_1(t), D_2(t), \dots, D_J(t)$.

2.2. BiLSTM

LSTM and BiLSTM are special structural variants of RNN, particularly, BiLSTM consists of two layers of LSTM with opposite directions, so it is called bidirectional LSTM. The interior structure of single layer LSTM is shown in the Fig. 2.

(1) Firstly, through the forgetting gate and decide what information to discard. The input of the forgetting gate is the output value h_{t-1} of the prior time and the information x_t of the new input. These two inputs are mapped to the [0,1] interval by activating the sigmoid function. The formula of forgetting gate is as follows:

$$f_t = \sigma(W_f * x_t + U_f * h_{t-1} + b_f) \quad (5)$$

where f_t is the state information processed by the forgetting gate; σ is the sigmoid activation function; W_f is the weight matrix between the input layer and the forgetting gate; U_f is the weight matrix between the hidden layer and the forgetting gate; h_{t-1} is the output of the time point of the previous hidden layer; x_t is the output at time t ; b_f is the bias vector of the forgetting gate.

(2) Secondly, through the input gate and decide what information to keep. Input the output value h_{t-1} of the prior time and the information x_t of the new input, through the activation function sigmoid, also map these two inputs to the [0,1] interval, representing the probability of retaining this part of information. The formula for the input gate is as follows:

$$i_t = \sigma(W_i * x_t + U_i * h_{t-1} + b_i) \quad (6)$$

At the same time, a new temporary state, i.e. internal memory unit c'_t , will be generated, and the old state c_{t-1} will be updated. The new state value c_t will be obtained by adding the new temporary state information with the old state information obtained through the forgetting gate. The formula of internal memory unit is as follows:

$$c'_t = \tanh(W_c * x_t + U_c * h_{t-1} + b_c) \quad (7)$$

$$c_t = f_t * c_{t-1} + i_t * c'_t \quad (8)$$

where i_t is the output of the input gate at time t ; W_i and U_i are the weight matrices of gate connection; b_i is the offset vector of

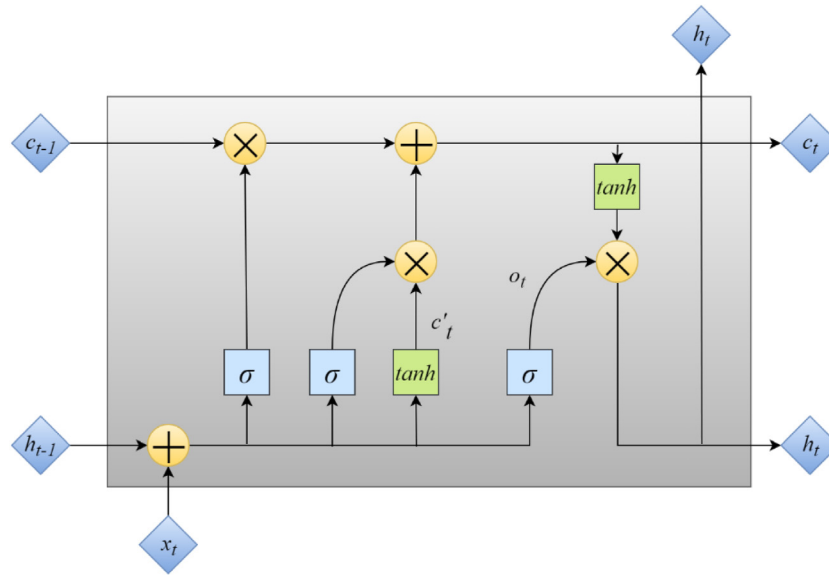


Fig. 2. One basic unit of LSTM.

the input gate; $\tanh$ is the activation function; W_c and U_c are the weight matrices of the calculation unit; b_c is the bias vector.

(3) Finally, the output h_t is obtained by multiplying the o_t of the input information processed by the sigmoid activation function and the unit state value transformed by the $\tanh$ activation function through the output gate. The formula of output gate is as follows:

$$o_t = \sigma(W_o * x_t + U_o * h_{t-1} + b_o) \quad (9)$$

$$h_t = o_t * \tanh(c_t) \quad (10)$$

where W_o and U_o are the connection weights of the output gate, b_o is the bias vector of the output gate, and h_t is the value of the output gate.

According to the LSTM described above, BiLSTM can be obtained by introducing double layer LSTM and setting the forward layer and the reverse layer. Sometimes our prediction may need to be decided by the previous information and the later information, that is to say, the value of a node at a certain time may be affected by the value near the time. The network structure is shown in the Fig. 3. We can see that the forward layer and the backward layer jointly connect with the output layer, which includes six shared weights $W_1 - W_6$.

The forward and back propagation layers calculate from time 1 to time t and from time t to time 1 respectively, and retain their respective states at each time. Lastly, the final output is acquired by combining the output values at the corresponding time of forward and backward layers at each time, which is expressed by mathematical expression as follows:

$$h_t = f(W_1 * x_t + W_2 * h_{t-1}) \quad (11)$$

$$h'_t = f(W_3 * x_t + W_5 * h'_{t+1}) \quad (12)$$

$$O_t = g(W_4 * h_t + W_6 * h'_t) \quad (13)$$

2.3. Attention mechanism

Attention mechanism is a kind of sequential processing optimization component. The core idea of attention mechanism is to focus more attention on important information and reduce the attention to other information. The internal structure of attention mechanism unit is shown in the Fig. 4. In order to extract effective information, attention mechanism is introduced into the deep

learning model, which is an effective probability weight allocation mechanism. The mathematical formula of attention mechanism in this paper is as follows:

(1) The correlation between hidden layer states h_j and h_i in each group was calculated.

$$e_{ij} = \tanh(W_1 * h_i + W_2 * h_j + b) \quad (14)$$

where e_{ij} represents the relationship between the i th state and the j th state, W indicates the weight, b is the offset vector.

(2) Using the softmax formula to convert the value of each correlation into a probability weigh.

$$a_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_j \exp(e_{ij})} \quad (15)$$

where a_{ij} is the attention weight of j to i , and $\sum_j a_{ij} = 1$.

(3) The probability of each input feature is multiplied by the hidden layer state of the corresponding input feature to represent the contribution of the feature to the predicted value, and the contribution of all input features is added as the output of the attention mechanism layer.

$$H_i = \sum_j a_{ij} * h_j \quad (16)$$

where H_i represents the final output state after passing through the attention layer. We apply the formula above to compute the corresponding weight of the coding sequence, and allocate the weight to each code. After that we will get a new coding sequence with enhanced function.

2.4. CNN

Convolutional neural network can extract global spatial features, reduce the number of parameters and features, and avoid over fitting. A regular CNN frame has several layers: input layer, convolution layer, pool layer, full connected layer and output layer. The detailed framework of CNN is as follows:

(1) Convolution layer

Convolution layer captures spatial features from crude oil futures prices at adjacent points by convolution operation. The parameters of convolution layer are generally composed of kernel number, kernel length and a group of neurons. In the forward propagation process, the convolution layer calculates the dot

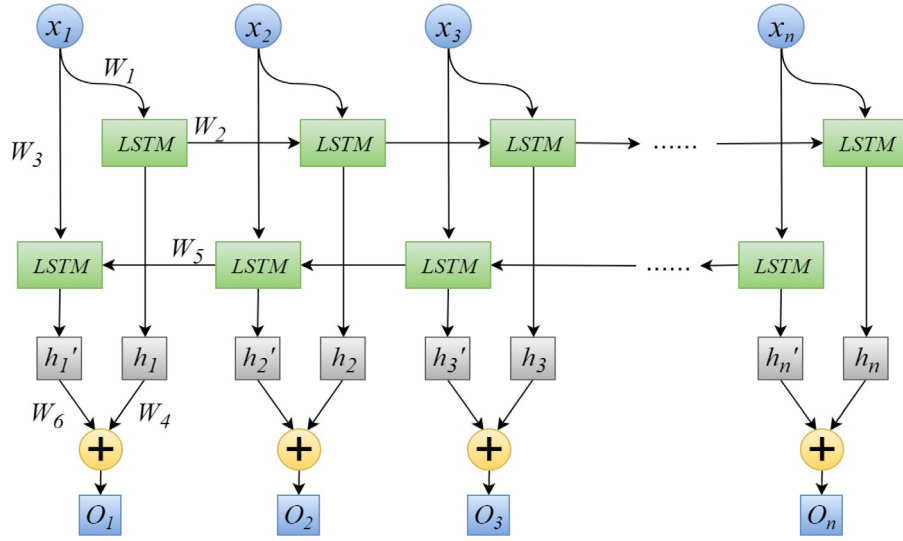


Fig. 3. Bidirectional LSTM model structure.

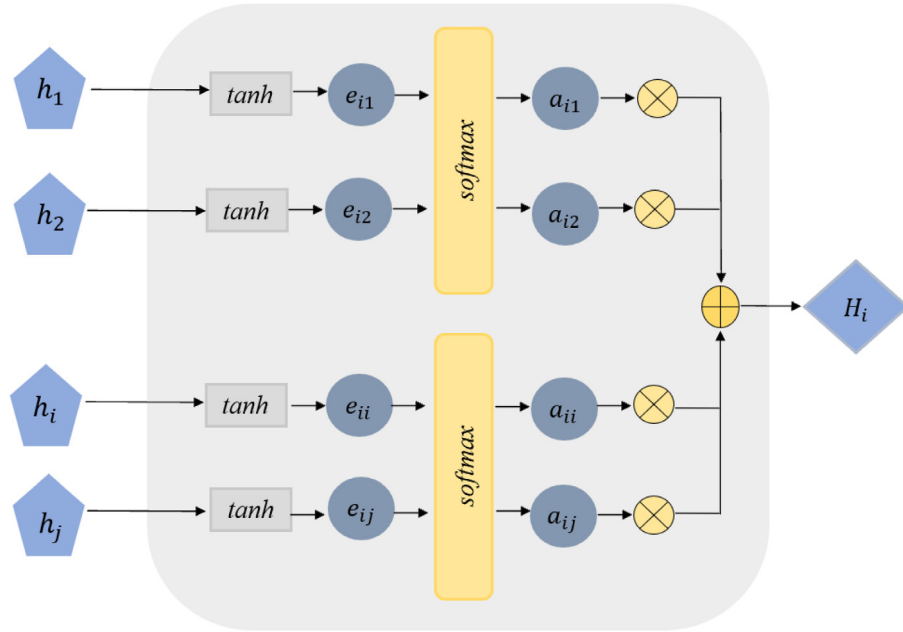


Fig. 4. One basic unit of Attention.

product between each kernel and the input data, the specific operation is expressed by the following formula:

$$x_j^l = f\left(\sum_{i \in M_j} x_i^{l-1} * k_{ij}^l + b_j^l\right) \quad (17)$$

where x_j^l is the j th output of layer l , M_j is the selected input maps, x_i^{l-1} is the i th output of layer $l-1$, k_{ij}^l is the weight between the i th output and the j th output, b_j^l is the bias vector, $*$ is the convolution operation, and f is the activation function.

(2) Pooling layer

Pooling layer is applied to decrease the number of parameters (such as the number of filters and training weights) and redundant features, which can be achieved by calculating the average or maximum value of a given area in feature maps. In addition, pooling layer can also be applied to control the convergence of neural networks (e.g. avoiding over fitting). The most common pooling

operation is maximum pooling, which selects the maximum value for the portion covered by the pooling filter.

(3) Fully connected layer

The fully connected layer can map the feature maps produced by convolution layer and pooling layer to another feature space. It also can extract and integrate useful information. The specific expression is as follows:

$$x^l = f(k^l x^{l-1} + b^l) \quad (18)$$

$$f(y) = \max(0, y) \quad (19)$$

where x^l is the final output vector, k^l is the weight between layer $l-1$ and layer l , x^{l-1} is the input vector, b^l is the bias vector, and f is the activation function. Here, we choose the Rectified Linear Unit (ReLU) as the activation function, which only activates positive values. Using this function can effectively avoid over fitting.

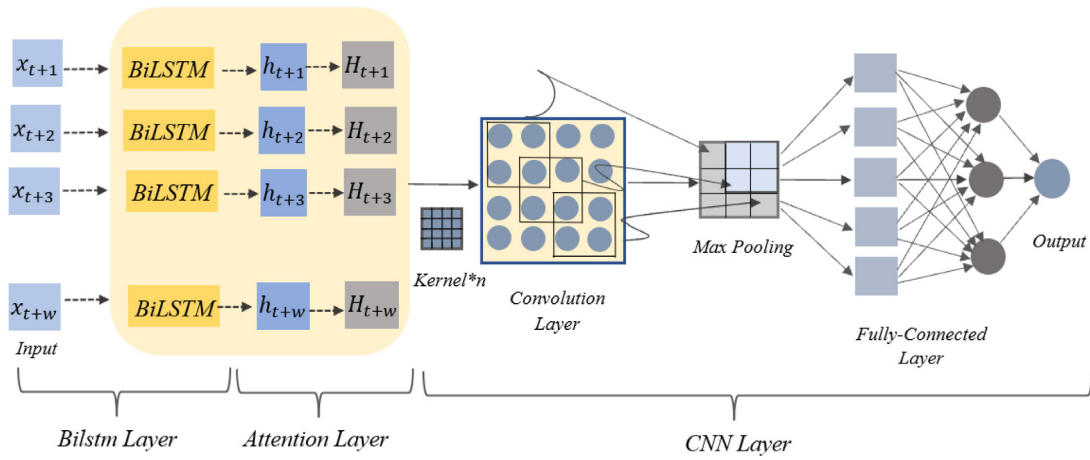


Fig. 5. BiLSTM-Attention-CNN framework.

2.5. Proposed method

In general most financial time series can be regarded as unstable signals, including WTI and Brent crude oil futures [43]. This paper proposes the novel decomposition-ensemble method. Firstly, use wavelet transform to decompose WTI and Brent crude oil futures daily closing price time series into main signal A_j and some noise signals $D_i (i = 1, \dots, J)$. Then, using A_j and D_i as the sample training, $J + 1$ BiLSTM-Attention-CNN models are built for sample training, and then A_j and D_i are predicted respectively. Finally, the linear combination of the predicted values to calculate the final error and prediction accuracy. The Fig. 6 shows the modeling process of daily closing price prediction of WTI and Brent crude oil futures. The detailed steps are as follows.

Step 1: Data decomposition using Wavelet Transform

The original time series $x(t)$ are imported and decomposed into certain several levels to get a low-frequency main signal A_j and several high-frequency noise signals $D_i (i = 1, \dots, J)$.

Step 2: Forecasting model using BiLSTM-Attention-CNN

The components obtained by Wavelet Transform are trained and predicted by BiLSTM-Attention-CNN model, and the predicted results are $A'_j(t)$ and $D'_i(t) (i = 1, \dots, J)$. The BiLSTM-Attention-CNN (we abbreviated it as BLAC) network structure is shown in Fig. 5. The windows size is w .

(1) Inputting the obtained subsequence into BLAC model, first obtain the temporal features through BiLSTM. After BiLSTM processing, a new coding sequence will be acquired. However, because of the huge size of time series, the more data will enter the BiLSTM layer, which will cause the effect of this layer to become worse. That is to say, the quality of the memory processing information in each BiLSTM cell will be degraded. Therefore, we introduce attention mechanism.

(2) Loading the attention mechanism into BiLSTM. Then allocate the probability weight of the new coding sequence at different time points through attention mechanism to enhance the features of the new coding sequence.

(3) Finally, the convolution layer unit is connected with the output of attention layer, and each convolution layer unit has the same weight connection matrix (that is to say, the shared weight), which reduces the parameters of CNN and makes the CNN model easier to train. The pooling layer is applied to down sample the output feature maps, which could integrate crucial features extracted from the convolution layer. The flat layer transforms these feature maps into one-dimensional vectors and obtains an output result through the full connection layer.

Step 3: Ensemble prediction

Finally, the prediction values of each component are reconstructed into the final prediction results $x'(t) = \sum_{i=1}^J D'_i(t) + A'_j(t)$.

The BLAC algorithm flow is shown in the Algorithm 1.

2.6. Evaluation criteria

To evaluate the prediction effect, this paper adopts the following four criteria to calculate the prediction error and accuracy, which include root mean square error (RMSE), mean absolute error (MAE), average absolute percentage error (MAPE), and determination coefficient (R^2). The prediction performance of the model increases with the decrease of MAE, RMSE and MAPE, and increases with the increase of R^2 . The formula of the above evaluation indexes is as follows:

(1) Root Mean Square Error (RMSE)

RMSE represents the standard deviation of the residual error between the predicted value and the observed value. The mathematical expression is as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{x}_i - x_i)^2} \quad (20)$$

(2) Mean Absolute Error (MAE)

MAE is a measure of the difference between two continuous variables and calculates the mean value of all absolute errors. The mathematical expression is as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{x}_i - x_i| \quad (21)$$

(3) Average Absolute Percentage Error (MAPE)

MAPE is often to judge the performance of predictions in statistics and more stable. The mathematical expression is as follows:

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{\hat{x}_i - x_i}{x_i} \right| \quad (22)$$

(4) Determination Coefficient (R^2)

R^2 represents the quality of a fitting by the change of data. It is in the interval of $[0,1]$, and the closer it is to 1, the better the fitting degree is. R^2 can be calculated by two

Algorithm 1: BiLSTM-Attention-CNNInput: Daily closing price time series $x(t)$ of WTI and Brent crude oil futures.Output: Evaluating indicator ($RMSE, MAE, MAPE, R^2$) of the forecasted data

```

1. Function Split dataset( $x, y$ )
2.     # 70% training and 30% testing data
3.     size ← length(series)*0.7
4.     train ← series[0...size]
5.     test ← series[size...length(size)]
6.      $x \leftarrow \text{train}$ 
7.      $y \leftarrow \text{train}-x$ 
8. Return  $x, y$ 
9. Function BLAC_model
10.    set random.seed(7)
11.    model = Sequential ( )
12.    model.add (Bidirectional(LSTM(neurons,return_sequences=True)))
13.    model.add (SeqSelfAttention(attention_activation='sigmoid', name='Attention'))
14.    model.add (Convolution1D(filter,filter_length))
15.    model.add (MaxPooling1D(pool_length))
16.    model.add (Flatten( ))
17.    model.add (Dense(1))
18.    model.compile (loss='mean_squared_error',optimizer='adam')
19.    for each  $i$  in range (epoch) do
20.        model.fit (x,y,epoch=1,shuffle=False)
21.        model.reset_states ( )
22.    end for
23. Return BLAC_model
24. Function BLAC_model_forecast (predictions, expected)
25.    epoch ← 1
26.    neurons ← 64
27.    predictions ← empty
28.    BLAC_model = fit_BLAC (test,epoch,neurons,filter,filter_length,pool_length)
29.    # Walk-forward validation on the test data
30.    for each  $i$  in range (length (test) ) do
31.        # make one-step forecast
32.         $x \leftarrow \text{test}[i]$ 
33.         $\hat{y} \leftarrow \text{forecast\_BLAC}$  (BLAC_model,  $x$ )
34.        # record forecast
35.        predictions.append( $\hat{y}$ )
36.        expected ← test[ $i$ ]
37.    end for
38. Return predictions, expected
39. Function Evaluating indicator(expected, predictions)
40.    Error ←  $RMSE, MAE, MAPE, R^2$ 
41. Return Error

```

parts (sum of squares of the expression (SSR) and total of sum of squares (SST)). The mathematical expression is as follows:

$$SSR = \sum_{i=1}^n (\hat{x}_i - \bar{x}_i)^2 \quad (23)$$

$$SST = \sum_{i=1}^n (x_i - \bar{x}_i)^2 \quad (24)$$

$$R^2 = \frac{SSR}{SST} \quad (25)$$

In the above formula, $\hat{x}_i$ is the predicted value, x_i is the true value, $\bar{x}_i$ represents the mean value of the true value.

Furthermore, the modification of DM test was used to ascertain whether there is obvious difference between the two models, that is, whether the model proposed in this paper is distinctly superior than other models. The fundamental principle is: supposing that $\{y_t\}$ is the price sequence of assets, $\{\hat{y}_{it}, i = 1, 2\}$ are

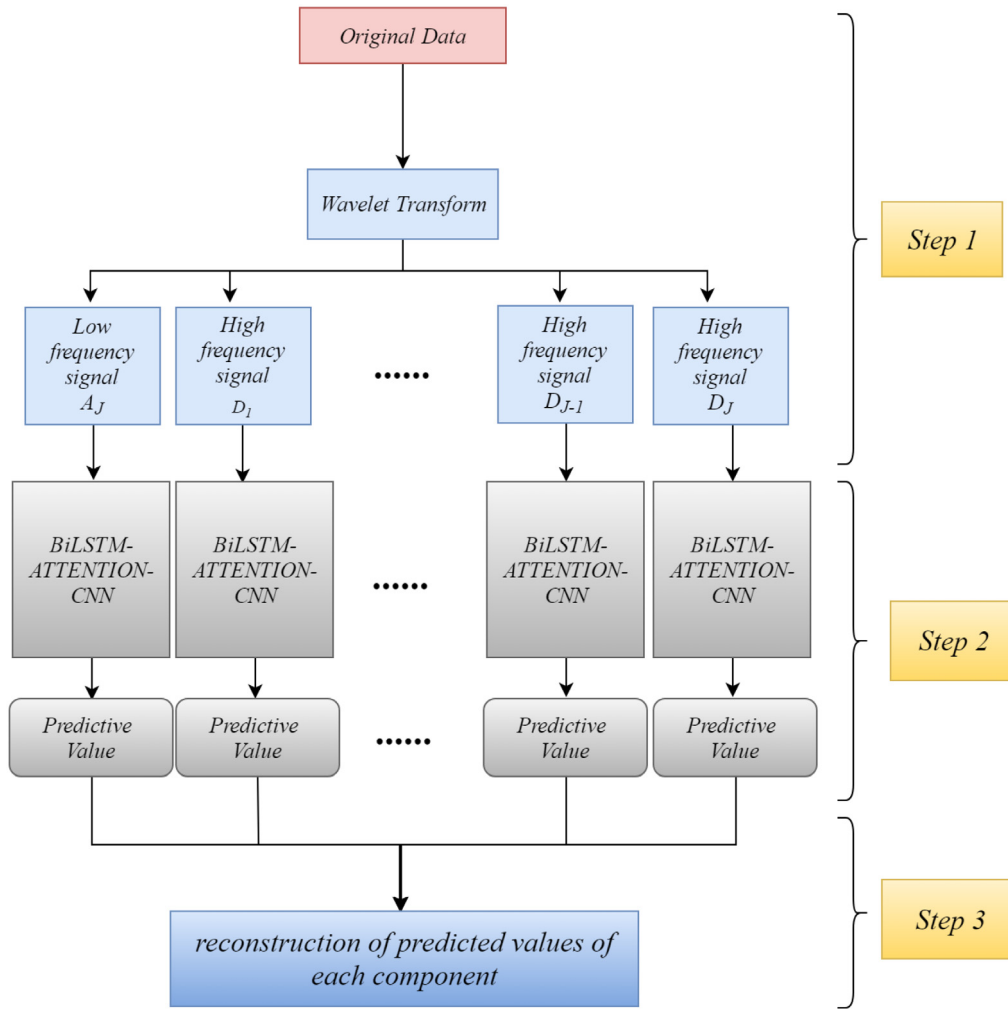


Fig. 6. The forecast flow chart based on WT-BLAC.

the predicted values of the two models, and $\{e_{it}, i = 1, 2\}$ are the prediction error of the two models:

$$e_{it} = y_t - \hat{y}_{it} \quad i = 1, 2 \quad (26)$$

Simply, the loss function is defined as follows:

$$Loss_i = L(y_t, \hat{y}_{it}) = L(e_{it}) \quad i = 1, 2 \quad (27)$$

where the loss function predicted by the i th model is expressed by $Loss_i$. And d_t is defined as a loss differential sequence with stable covariance but short memory:

$$d_t = L(e_{1t}) - L(e_{2t}) \quad t = 1, 2, \dots, T \quad (28)$$

$$H_0: E(d_t) = 0 \quad t = 1, 2, \dots, T \quad (29)$$

where T represents the length of the tested sequence. The null hypothesis H_0 of the test is that the two models have the same prediction ability and the expected value is equal. In order to better predict the long memory sequence, ω^2 is used to represent the long memory variance of d_t .

$$\sqrt{T}(\bar{d} - E(d_t)) \xrightarrow{d} N(0, \omega^2) \quad (30)$$

where $\bar{d}$ is the $T^{-1} \sum_{t=1}^T d_t$, ω^2 means the $\sum_{j=-\infty}^{\infty} \gamma_j$ and γ_j represents $Cov(d_t, d_{t-j})$. Moreover, ω^2 was advanced to represent a more consistent long-run variance indicator, $\hat{\omega}^2$ was expressed:

$$\hat{\omega}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{h-1} \hat{\gamma}_j \quad (31)$$

where $\hat{\gamma}_j$ is $T^{-1} \sum_{t=j+1}^T (d_t - \bar{d})(d_{t-j} - \bar{d})$. In order to make higher accuracy between various sample size results applying to the test, the modification of DM statistics was expressed:

$$MDM = \sqrt{T+1-2h+T^{-1}h(h-1)}(\bar{d}/\hat{\omega}) \quad (32)$$

where the previous step of prediction represented by h . The core of MDM test is to replace the standard normal distribution of DM test with Student's t critical values. In this paper, MSE, MAE and MAPE are used as basic loss functions of MDM test.

3. Analysis of experimental results

3.1. Data description and processing

The original price data of WTI and Brent crude oil futures which derived from investing (<https://cn.investing.com>). The main crude oil futures markets include North Sea Brent crude oil (Brent) of the London International Petroleum Exchange and West Texas Intermediate (WTI) of the New York Mercantile Exchange (NYMEX). The daily closing prices of WTI and Brent crude oil futures are collected from January 4, 2010 to July 31, 2020 (as shown in Fig. 7). We select the data of trading day as the test data, and the data length of WTI and Brent crude oil futures is 2746 and 2732 respectively. Two groups of data are divided into two parts, this study chose the first 70% of series be the training part and the last 30% of series be the test part. Before sending series into the model, we should normalize each data into [0,1]

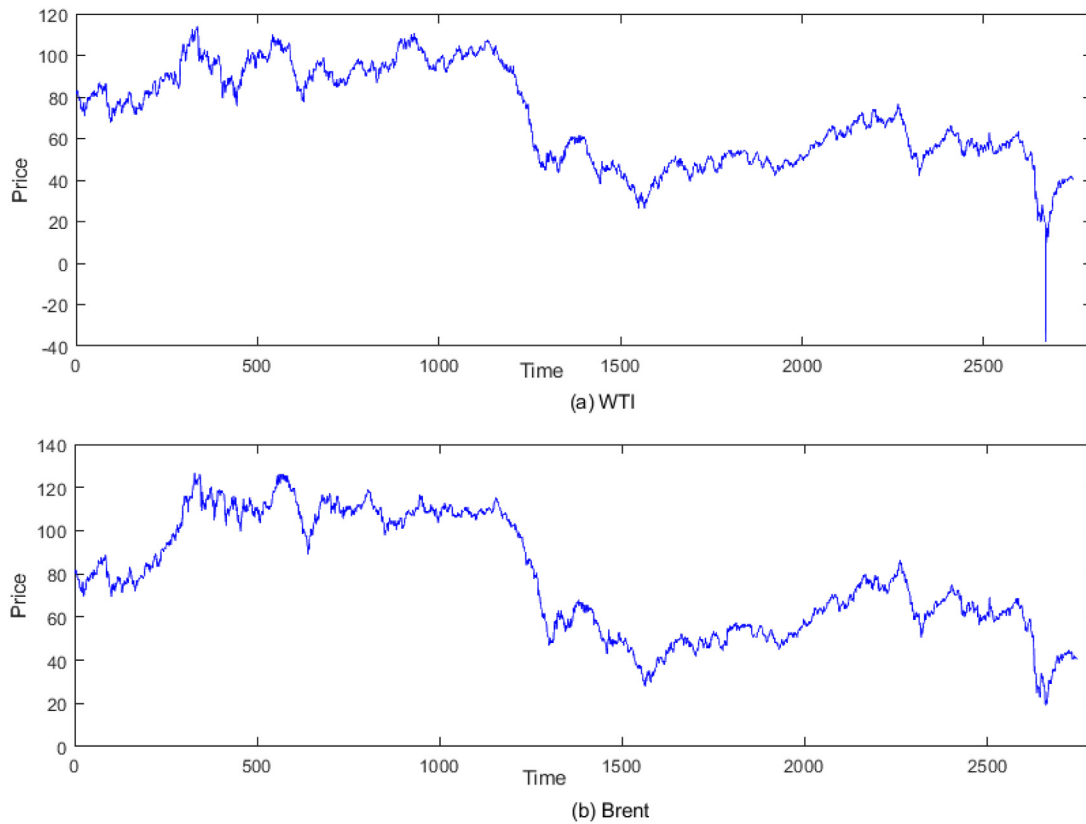


Fig. 7. Daily closing price of Brent and WTI crude oil futures.

interval, which will reduce the impact of noise, accelerate the updating speed of neural network parameters and improve the training performance of the model. The following formula is used for standardization:

$$\tilde{x}_t = (x_t - \min x_t) / (\max x_t - \min x_t) \quad (33)$$

where x_t and $\tilde{x}_t$ refers to the data before and after normalization respectively.

The descriptive statistical test values of daily closing prices of WTI and Brent crude oil futures are shown in Table 1. Both WTI and Brent crude oil futures prices show the significant characteristics of “leptokurtic”, “fat-tailed” and “right-skewed”. Meanwhile the Jarque–Bera statistics of two kinds of crude oil futures price time series show that the null hypothesis should be rejected at 1% significance level, indicating that both of them do not obey normal distribution. Accordingly, the null hypothesis of Ljung–Box test is that there is no autocorrelation in a certain time series. The Ljung–Box statistics of two kinds of crude oil futures price time series show that the null hypothesis should be rejected at 1% significance level, indicating that both of them have 10 order lag autocorrelation and long-memory.

Additionally, we also add ADF test to verify whether the two crude oil futures price sequences are non-stationary data. As shown in Table 2, the p -value of the two time series are very large, and the T statistics are greater than the T statistics under three confidence levels, which shows that we cannot reject the null hypothesis that the two sequences are both non-stationary time series.

Through the above statistical analysis, we found that two kinds of crude oil futures price time series can be viewed as non-stationary signals. Therefore, wavelet transform is applied to decompose and reconstruct the non-stationary signal into a linear combination of a low-frequency main signal and several noise

Table 1

The statistical analysis results of WTI and Brent crude oil futures price.

Index	WTI	Brent
Count	2746	2732
Mean	70.2089	77.9742
Standard deviation	22.9411	26.6013
Max	113.9300	126.6500
Min	−37.6300	19.3300
Skewness	0.0057***	0.0993***
Excess kurtosis	1.9342***	1.6973***
J-B	$1.30 \times 10^{2***}$	$1.98 \times 10^{2***}$
Q(10)	$2.68 \times 10^{4***}$	$2.69 \times 10^{4***}$

Note: J-B denotes the Jarque–Bera Statistics. Q(10) is the Ljung–Box statistic for up to 10th order serial correlation. *, ** and *** denote rejection of the null hypothesis at the 10%, 5%, 1% significance level respectively.

Table 2

ADF test results of WTI and Brent crude oil futures price series.

	T-statistic	Prob	1% T-statistic	5% T-statistic	10% T-statistic
WTI	−1.3187	0.6206	−3.4327	−2.8626	−2.5673
Brent	−1.0004	0.7531	−3.4327	−2.8626	−2.5673

high-frequency signals, which can reduce the impact of noise and improve the prediction accuracy.

The performance of wavelet transform usually depends on the choice of wavelet basis function. In this paper, Daubechies wavelet [44] is selected to fit the original time series for many reasons. Daubechies wavelet has good regularity, which makes the signal reconstruction process more smoothly [45] and it has been found to have good denoising effect in various studies [46–49]. For the above reasons, we choose db4 wavelet to decompose and reconstruct the original data into three levels. Fig. 8 shows the results of wavelet transform. One low-frequency main signal and three high-frequency noise signals are obtained respectively.

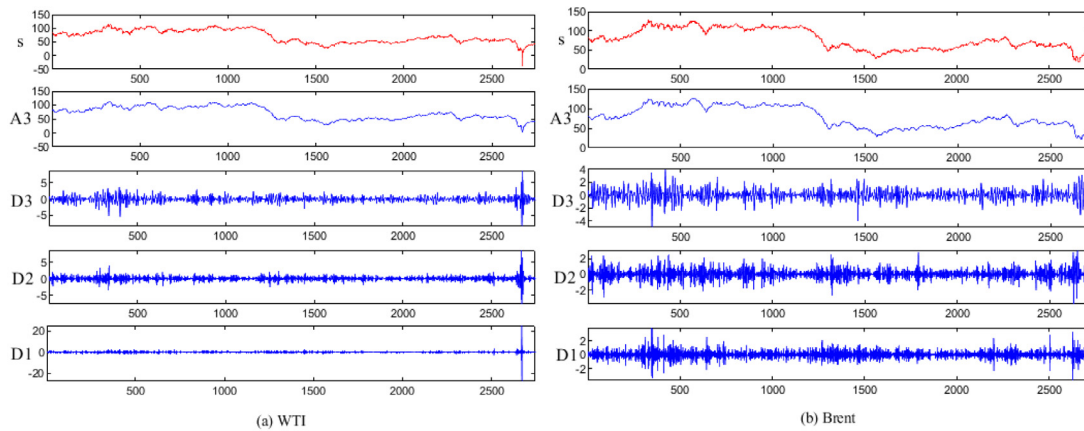


Fig. 8. Wavelet transform result of daily closing price of WTI and Brent crude oil futures.

So the original sequence $x(t)$ can be expressed as $x(t) = A_3 + D_3 + D_2 + D_1$. Where A_3 expresses the low-frequency component, D_3 , D_2 , D_1 are the high-frequency components of different frequencies. The trend of low-frequency component A_3 is roughly the same as that of the original data $x(t)$, and D_3 , D_2 , D_1 fluctuate around zero, indicating that the noise part of the original sequence has been well separated. Then they are predicted by the combination model proposed in this paper respectively.

3.2. Hyperparameter selection

In this study, after repeated experiments, we found the optimal value of the model's hyperparameters. In BLAC model, the number of neurons in BiLSTM, Epoch, length of sliding window, Batch size, Dropout, length and number of convolution kernel play the important role in the prediction. The influence and significance of each hyperparameter on the model are explained as follows:

Batch size: it is used to find the best balance between memory efficiency and memory capacity when the data set is too large. Its size will affect the model optimization degree and training speed. Increase batch size, the amount of data read in at one time increases, the required memory capacity increases, and the relative processing speed increases. Appropriate batch size, which can increase the accuracy of the gradient descending direction, reduce the training shock and accelerate the convergence.

Epoch: it is not enough to transfer the complete data set once in the neural network, and we need to transfer the complete data set many times in the same neural network. Therefore, it is not enough to update only once or use one epoch. As the number of epoch increases, the number of updates of weights in the neural network also increases, and the curve changes from under fitting to over fitting. The final accuracy will not change as the number of epoch increases, and then we get a better value.

Windows: it represents the length of each input sequence.

BiLSTM units: generally speaking, the more neurons, the higher the accuracy rate. However, it is easy to have overfitting. At this time, dropout should be adopted.

Dropout: it will randomly and artificially discard some neurons during training, which will reduce the complex adaptive relationship between neurons: because dropout program causes two neurons not to appear in one dropout network every time. In this way, the updating of the weights is no longer dependent on the joint action of the hidden nodes of the fixed relationship, which prevents some features from being effective only under other specific features.

Convolution kernel: the $1 * 1$ convolution kernel is adopted in this paper, which can greatly increase the nonlinear characteristics (using the subsequent nonlinear activation function) under

Table 3

Hyper-parameters of sub-series of WT-BiLSTM-Attention-CNN.

Hyper-parameters	Number	Hyper-parameters	Number
Epoch	100	Convolution kernel	64
Window	3	Filter length	2
Batch size	80	Maxpooling length	2
BiLSTM units	64	Dropout	0.25

Note: The "Window" means the scale of input sequence. The "Epoch" means numbers in progress. "Dropout" is the rate of dropout of CNN at a certain probability. Batch size represents the number of training samples in each batch.

Table 4

Forecasting evaluation results of original series and A3 under four statistical performance indicators.

Data	WTI				Brent			
	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²
Original	2.6181	1.1656	2.7294	0.9545	1.6336	1.1777	2.2150	0.9841
A3	1.4485	1.0475	3.0002	0.9857	1.0847	0.7407	1.4858	0.9929

Notes: Numbers in bold mean that their corresponding model has the best forecasting performance among all models.

the premise of keeping the scale of the feature map unchanged (i.e. without losing the resolution), and make the network very deep. When the data is complex (the hidden features are difficult to extract), more convolution kernels are needed to improve the accuracy.

Finally, the filter length and maxpooling length are determined according to the dimensions encoded after processing through the attention mechanism, and their roles have been mentioned in 2.4.

The selection of hyperparameters is shown in Table 3.

Each component after wavelet transform is predicted by BLAC model. The prediction effect of low-frequency principal component and three high-frequency noise components is shown in the Fig. 9. The real value is represented by a solid blue line, while the predicted value is represented by a dotted red line. It can be found from the Fig. 9 that the predicted value of the low-frequency sequence is almost consistent with the actual value, and the error of prediction is obviously less than the prediction error of the original data.

As shown in Table 4, the fitting degree of A3 principal component of WTI by wavelet transform is as high as 0.98, far higher than 0.95 of the original sequence. This indicates that the influence of noise can be effectively reduced through wavelet transform, which makes the prediction effect more accurate. High frequency series represent volatility term, which includes periodic term and random term. In practical time series problems

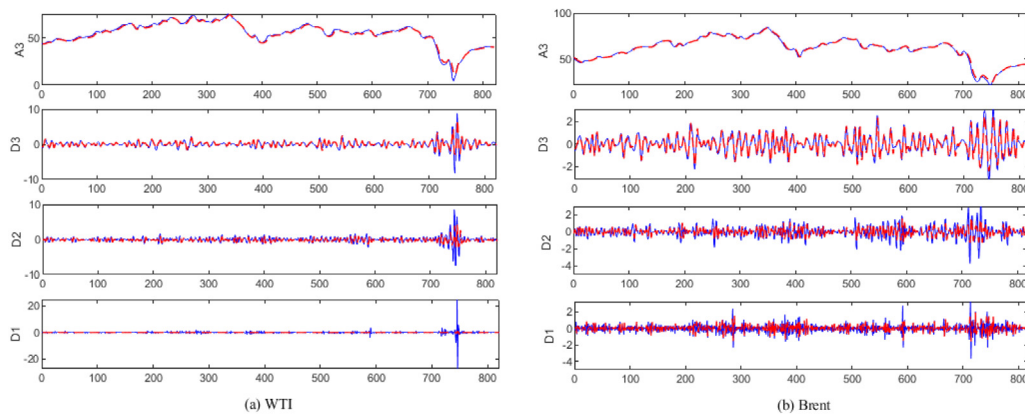


Fig. 9. Forecasting results of subseries for daily closing price of WTI and Brent crude oil futures.

Table 5

Forecasting evaluation results under four statistical performance indicators.

Models	WTI				Brent			
	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²
BiLSTM	2.6928	1.2849	2.9770	0.9519	1.7186	1.3041	2.3963	0.9824
BLAC	2.6181	1.1656	2.7294	0.9545	1.6336	1.1777	2.2150	0.9841
CNN-LSTM	4.1226	2.2733	6.1430	0.8872	3.7575	2.1827	5.2557	0.9160
CNN	5.3265	4.1991	9.1029	0.8117	4.0692	2.7261	6.2662	0.9015
CNN-BiLSTM	4.8437	2.4861	7.3427	0.8443	3.5752	2.2134	5.1565	0.9240
GRU	2.8336	1.4153	3.2947	0.9467	1.7657	1.2726	2.4771	0.9815
LSTM	2.9619	1.5572	3.6216	0.9418	2.1963	1.6489	3.2725	0.9713
BiLSTM-CNN	3.4235	2.4637	4.8848	0.9222	1.9098	1.2681	2.6778	0.9783
LSTM-CNN	3.6368	2.4733	5.4053	0.9122	2.1606	1.5404	3.0925	0.9722
BP	2.9653	1.4142	3.5639	0.9416	2.1809	1.5196	3.1735	0.9717
SVR	5.6715	3.3630	9.3963	0.7865	5.6269	4.8044	9.7157	0.8117
WT-BLAC	2.2518	1.1780	2.6261	0.9663	1.2936	0.8894	1.8027	0.9900

Notes: Numbers in bold mean that their corresponding model has the best forecasting performance among all models.

such as financial data, high frequency information component is often random factor, which can be considered as insensitive to historical data.

3.3. Forecasting performance

First, we compared the prediction performance of WT-BLAC with other models under different evaluation indicators. Fig. 10 shows the prediction results of all models. The true value is indicated by the blue line, while the red line and the purple line expresses the predicted value of BLAC model after wavelet transform and the predicted value of BLAC model respectively. Table 5 shows the prediction error results of different models based on the four evaluation indicators (RMSE, MAPE, MAE and R²).

Next, we used the MDM test to verify whether WT-BLAC was statistically significantly superior to other models. Table 6 lists the results of the MDM test. We can draw the following conclusions:

(1) From the error results in Table 5, we can see that the model WT-BLAC has the highest fitting degree for the forecasting of WTI and Brent crude oil futures price, R² of WTI is 0.9663, and R² of Brent is 0.9900. Their RMSE, MAE and MAPE are the smallest among all models. Moreover, in the model without decomposition method, our combined model BiLSTM-Attention-CNN is superior to other compared models in prediction accuracy.

(2) After wavelet transform, the two crude oil futures prices are predicted by our BLAC model. It can be found that the coincidence between the prediction results and the original data is significantly improved. From the enlarged part of Fig. 10, it can be seen that the prediction effect of abnormal data fluctuation

after wavelet transform is obviously better than that of using our combined model BLAC before decomposition.

(3) For the WTI forecast, on April 20, 2020, the final delivery price of the WTI crude oil futures contract (due on April 21 and delivered in May) on the Chicago Mercantile Exchange (CME) fell to - 37.63 dollars/barrel, a decrease of more than 300%. This is the first time that the WTI crude oil futures price fell to a negative value, which is an event that we could not anticipate in advance. This further confirms that the time series of crude oil futures are unstable and difficult to predict accurately. For Brent's prediction, as two major crude oil futures products worldwide, Brent crude oil futures price has also been affected, and the corresponding drop in the next day is a large margin, but we can predict it more accurately through WT-BLAC model.

(4) In addition, we analyze traditional deep learning models such as CNN, LSTM and BiLSTM. The DL model performs better than the ML model (on the basis of the smaller values of RMSE, MAE, and MAPE). The cause might be that DL model has better generalization capabilities, especially after accepting a lot of WTI and Brent crude oil futures price data. Particularly, our BiLSTM-Attention-CNN combination model is superior to other models, and the accuracy of the model can be further improved after wavelet transform. In contrast, for example, as shown in Table 6, RMSE, MAE and MAPE of SVR are 5.6282, 4.8047 and 9.7151 respectively in daily closing price prediction of Brent crude oil futures, and values of these indicators are the largest among all methods. As shown by the light purple line in Fig. 10, we can see that SVR model can predict the overall trend, but there is a big error in the prediction of daily price of two crude oil futures.

(5) By comparing BiLSTM-CNN model with CNN-BiLSTM model, we find that BiLSTM-CNN model is invariably better than

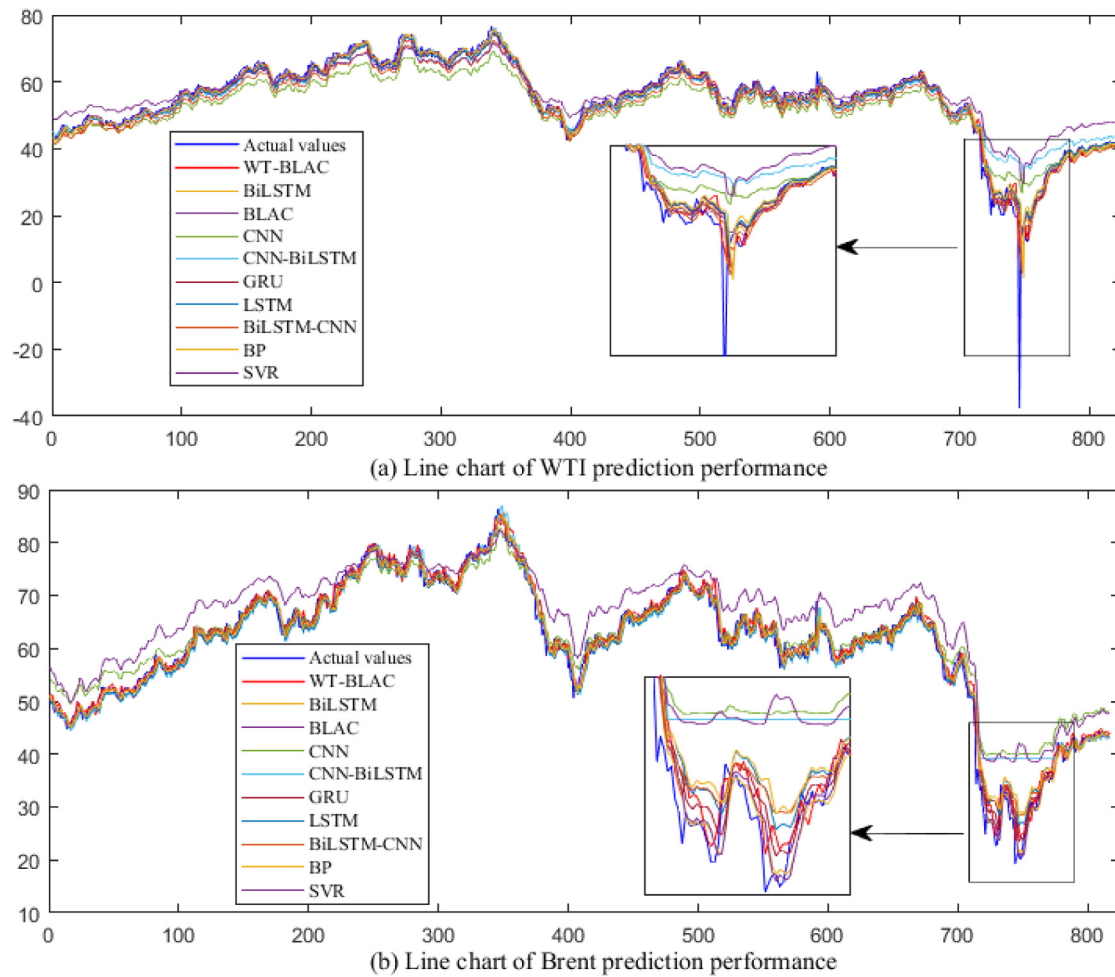


Fig. 10. Forecast results of two kinds of crude oil futures through all models.

CNN-BiLSTM model according to three performance indicators (RMSE, MAE and MAPE). These results seem to indicate that the different stacking order of CNN and BiLSTM will lead to different prediction accuracy. BiLSTM-CNN has a good performance, probably because the initial layer BiLSTM acts as an encoder. As long as data has been processed, it will record the daily WTI and Brent crude oil futures price series. Specifically, for each input, there is an output that stores not only the primal information, but also the processed information. Next, the CNN layer extracts local features by using the processed new coding sequence data to improve the prediction accuracy. Correspondingly, the convolution layer, the first layer of CNN-BiLSTM, will disturb our daily WTI and Brent crude oil futures price series to a certain extent when extracting data features. To sum up, if a set of data with order is scrambled and input into the BiLSTM layer, then this model is meaningless. This is because BiLSTM cannot play its advantages and just act as another fully connected layer.

(6) Consequences of the MDM test are expressed in Table 6. For the proposed model and other compared models, the results of the MDM test based on the MSE, MAE and MAPE loss functions represent the rejection of the null hypothesis at the 1% significance level in 26 cases, at the 5% significance level in 2 cases and at the 10% significance level in 2 cases for panel A. In term of panel B, all the cases indicated that the null hypothesis was rejected at the 1% significance level. Therefore, the results confirm that the loss function errors are statistically significant among the proposed model and other compared models which also suggest that WT-BLAC has better predictive ability of accuracy than other compared models.

3.4. Robustness check

The empirical results in Section 3.3 confirm the strong superiority of the prediction performance of the BLAC combined model with wavelet transform proposed in this paper. In this section, we will further verify the robustness of the model. This robustness check is carried out through the following two aspects: firstly, we check whether the model proposed in this paper still obtains superior results under the conditions of different training sets and test sets. Secondly, we examine the prediction performance of the model proposed in this paper in the recent conflict between Russia and Ukraine.

3.4.1. The proportion of training set and test set

We conducted the experiment using a shorter training set, selecting the first 40% of the observed values as the training set and the last 60% of the original sequence as the test set. MDM test is still used as the evaluation standard. The results of MDM test based on the three loss functions are shown in Table 7.

For WTI, the final MDM test results based on three loss functions show the rejection of the null hypothesis at the 1% significance level in all cases. Meanwhile, the results show that for Brent, only one case did not reject the null hypothesis. The results also suggest that loss function reductions are statistically significant for the proposed model and the related models. The proposed model WT-BLAC still has better prediction ability than other models.

Table 6

Forecasting evaluation results of MDM test between WT-BLAC model and other models. (Training set: Test set = 7:3).

Models	MSE	MAPE	MAE	MDM_1	MDM_2	MDM_3
Panel A: WTI						
BiLSTM	0.0359**	$1.67 \times 10^{-10***}$	$8.49 \times 10^{-21***}$	-2.1009	-6.4713	-9.6114
BLAC	0.2349	0.4518	0.7549	-1.1885	-0.7526	0.3122
CNN	$7.69 \times 10^{-13***}$	$1.22 \times 10^{-52***}$	$1.86 \times 10^{-183***}$	-7.2823	-16.4273	-38.0730
CNN-LSTM	$5.09 \times 10^{-5***}$	$1.76 \times 10^{-17***}$	$5.04 \times 10^{-34***}$	-4.0730	-8.7021	-12.7340
CNN-BiLSTM	$1.21 \times 10^{-5***}$	$4.82 \times 10^{-15***}$	$1.42 \times 10^{-31***}$	-4.4021	-7.9827	-12.1991
GRU	0.0774*	$4.83 \times 10^{-6***}$	$4.84 \times 10^{-10***}$	-1.7678	-4.6023	-6.3000
LSTM	0.0359**	$1.67 \times 10^{-10***}$	$8.49 \times 10^{-21***}$	-2.1009	-6.4713	-9.6114
LSTM-CNN	$3.33 \times 10^{-5***}$	$3.80 \times 10^{-33***}$	$4.50 \times 10^{-115***}$	-4.1723	-12.5443	-26.9512
BiLSTM-CNN	$2.56 \times 10^{-7***}$	$8.55 \times 10^{-47***}$	$1.49 \times 10^{-124***}$	-5.1969	-15.3234	-28.4748
BP	0.0548*	$9.87 \times 10^{-6***}$	$2.11 \times 10^{-7***}$	-1.9226	-4.4477	-5.2337
SVR	$2.82 \times 10^{-9***}$	$3.80 \times 10^{-27***}$	$3.14 \times 10^{-56***}$	-6.0077	-11.1890	-17.0874
Panel B: Brent						
BiLSTM	$1.15 \times 10^{-8***}$	$2.48 \times 10^{-9***}$	$1.04 \times 10^{-23***}$	-5.7660	-6.0300	-10.3590
BLAC	$4.93 \times 10^{-6***}$	$1.69 \times 10^{-5***}$	$1.41 \times 10^{-14***}$	-4.5983	-4.3275	-7.8396
CNN	$1.41 \times 10^{-19***}$	$3.78 \times 10^{-31***}$	$1.48 \times 10^{-58***}$	-9.2860	-12.1071	-17.5160
CNN-LSTM	$4.17 \times 10^{-14***}$	$9.28 \times 10^{-21***}$	$5.81 \times 10^{-41***}$	-7.6920	-9.6025	-14.1801
CNN-BiLSTM	$1.27 \times 10^{-4***}$	$4.66 \times 10^{-23***}$	$1.54 \times 10^{-49***}$	-7.8535	-10.1958	-15.8525
GRU	$9.62 \times 10^{-10***}$	$5.02 \times 10^{-16***}$	$9.85 \times 10^{-28***}$	-6.1881	-8.2797	-11.3286
LSTM	$1.49 \times 10^{-22***}$	$2.52 \times 10^{-38***}$	$2.99 \times 10^{-64***}$	-9.8105	-13.6427	-18.5301
BiLSTM-CNN	$2.35 \times 10^{-10***}$	$7.32 \times 10^{-15***}$	$6.77 \times 10^{-24***}$	-6.4172	-7.9278	-10.4055
LSTM-CNN	$7.90 \times 10^{-16***}$	$9.98 \times 10^{-32***}$	$2.25 \times 10^{-53***}$	-8.2211	-12.2357	-16.5698
BP	$9.49 \times 10^{-15***}$	$1.01 \times 10^{-24***}$	$4.65 \times 10^{-47***}$	-7.8929	-10.6083	-15.3792
SVR	$1.21 \times 10^{-64***}$	$1.23 \times 10^{-94***}$	$2.82 \times 10^{-214***}$	-18.5992	-23.6583	-43.4130

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5%, 1% significance level respectively. MDM_1 , MDM_2 and MDM_3 represent the value of MDM test statistics under the loss function of MSE, MAPE and MAE respectively. H_0 : WT-BLAC model has the same predictive power as other models representing the null hypothesis. H_1 : WT-BLAC model has better predictive ability than the other model representing the alternative hypothesis.

Table 7

Forecasting evaluation results of MDM test between WT-BLAC model and other models. (Training set: Test set = 4:6).

Models	MSE	MAPE	MAE	MDM_1	MDM_2	MDM_3
Panel A: WTI						
BiLSTM	$9.48 \times 10^{-127***}$	$2.44 \times 10^{-295***}$	0.0000***	-26.2024	-45.7367	-65.0756
BLAC	$1.07 \times 10^{-12***}$	$1.55 \times 10^{-50***}$	$1.07 \times 10^{-157***}$	-7.1777	-15.4758	-29.9557
CNN	$1.27 \times 10^{-265***}$	$2.13 \times 10^{-289***}$	0.0000***	-42.3510	-45.0580	-76.4505
CNN-LSTM	0.0000***	0.0000***	0.0000***	-65.8049	-61.6187	-115.9123
CNN-BiLSTM	0.0000***	0.0000***	0.0000***	-63.9913	-61.7068	-115.2342
GRU	$1.46 \times 10^{-208***}$	0.0000***	0.0000***	-35.8623	-59.9084	-85.9646
LSTM	$9.12 \times 10^{-228***}$	0.0000***	0.0000***	-38.0518	-56.2406	-87.0164
LSTM-CNN	$4.99 \times 10^{-146***}$	$9.05 \times 10^{-309***}$	0.0000***	-28.5605	-47.2758	-75.8102
BiLSTM-CNN	$1.78 \times 10^{-135***}$	0.0000***	0.0000***	-27.2792	-51.0103	-74.6696
BP	0.0000***	0.0000***	0.0000***	-60.7351	-61.8813	-110.8819
SVR	0.0000***	0.0000***	0.0000***	-53.8073	-53.7369	-102.3327
Panel B: Brent						
BiLSTM	$1.38 \times 10^{-153***}$	$1.13 \times 10^{-237***}$	$8.37 \times 10^{-255***}$	-29.4829	-39.2128	-41.1621
BLAC	0.1158	$1.34 \times 10^{-5***}$	$2.63 \times 10^{-9***}$	-1.5736	-4.3669	-5.9864
CNN	$5.09 \times 10^{-208***}$	$5.21 \times 10^{-279***}$	0.0000***	-35.8281	-43.9206	-60.2654
CNN-LSTM	$3.93 \times 10^{-260***}$	0.0000***	0.0000***	-41.7686	-53.8353	-65.0686
CNN-BiLSTM	$1.68 \times 10^{-213***}$	$2.12 \times 10^{-306***}$	0.0000***	-36.4557	-47.0597	-59.1087
GRU	$6.84 \times 10^{-163***}$	$4.33 \times 10^{-250***}$	$2.33 \times 10^{-269***}$	-30.5885	-40.6257	-42.8197
LSTM	$8.79 \times 10^{-175***}$	$1.04 \times 10^{-269***}$	$1.54 \times 10^{-303***}$	-31.9859	-42.8597	-46.7306
BiLSTM-CNN	$6.49 \times 10^{-130***}$	$4.54 \times 10^{-218***}$	$1.57 \times 10^{-247***}$	-26.6072	-36.9779	-40.3344
LSTM-CNN	$5.46 \times 10^{-143***}$	$1.24 \times 10^{-230***}$	$1.66 \times 10^{-286***}$	-28.2082	-38.4111	-44.7771
BP	$1.06 \times 10^{-305***}$	0.0000***	0.0000***	-46.9790	-60.3440	-76.8877
SVR	$4.44 \times 10^{-268***}$	$8.28 \times 10^{-315***}$	0.0000***	-42.6738	-48.0291	-64.6616

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5%, 1% significance level respectively. MDM_1 , MDM_2 and MDM_3 represent the value of MDM test statistics under the loss function of MSE, MAPE and MAE respectively. H_0 : WT-BLAC model has the same predictive power as other models representing the null hypothesis. H_1 : WT-BLAC model has better predictive ability than the other model representing the alternative hypothesis.

3.4.2. Recent dataset validation

According to the above analysis, the model proposed in this paper can well predict the impact of extreme events. Therefore, we specially added the data from August 2020 to August 2022 (including the impact of the recent war in Ukraine) to further verify the superiority of the model.

Fig. 11 shows the prediction comparison diagram of the model when the training set is 0.4 and the test set is 0.6. The MDM test again proves that the model proposed in this paper has excellent prediction performance.

As can be seen from the Fig. 11, starting from the 200th coordinate of the test set, the prices of the two crude oil futures

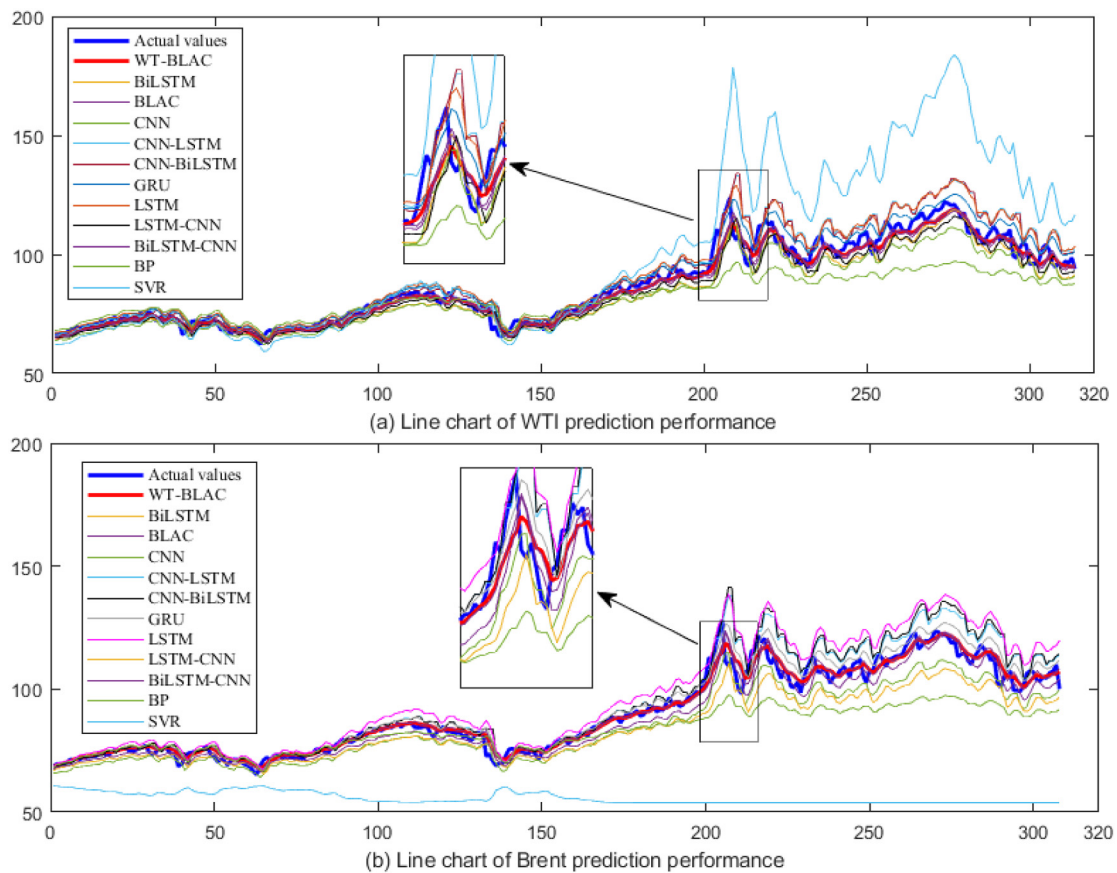


Fig. 11. Forecast results of two kinds of crude oil futures through all models.

began to rise, corresponding to the Russian Ukrainian conflict that broke out at the end of February 2022. On March 1, the daily closing price of WTI crude oil futures rose from \$95.72/barrel on February 28 to \$103.41/barrel, which was the first time in recent years that it broke through the \$100 level and remained high. In the meantime, Brent crude oil futures were also affected.

The bold red line and blue line in Fig. 11 respectively represent the prediction curve and the true value curve of the model proposed in this paper. We can intuitively see that the trend of these two curves is almost the same, and according to Table 8 and Table 9, the MDM test also confirms that the prediction effect of the proposed model is still better than other models under different proportions of training sets and test sets.

4. Conclusion

Crude oil plays an indispensable role in the global financial market. In particular, WTI and Brent are the global crude oil pricing benchmarks. However, in recent years, the situation has changed, that is, the application scope of the two crude oil futures is more extensive, and the price of crude oil futures is more sensitive to geographical and other globalization factors. Therefore, this study proposes a novel hybrid BiLSTM with wavelet transform combinatorial model to predict the daily closing prices of WTI and Brent crude oil futures. Our BiLSTM-Attention-CNN model is composed of BiLSTM unit, Attention mechanism and CNN component. BiLSTM unit can deal with long-term and long-term dependence problem. The experimental results indicate that bidirectional LSTM is better than LSTM. Attention mechanism is able to enhance the weight of each data in the sequence, while CNN unit can extract the spatial features of data.

To appraise the performance of the proposed model, we have made lots of experiments on the daily closing price data of two crude oil futures. The evaluation indexes are RMSE, MAPE, MAE and R^2 . From the empirical results, we figure out that the WT-BiLSTM-Attention-CNN model has the greatest prediction effect among different prediction models. In addition, RMSE, MAPE and MAE are the lowest among all the evaluation indexes, and R^2 is the highest, which indicates that proposed model can be adopted as a particularly advanced method for the prediction of crude oil futures price. Moreover, MDM test further verifies the superiority of WT-BiLSTM-Attention-CNN model. Finally, through robustness check, changing the proportion of different training sets and test sets, and additional analysis of recent data sets, we prove that the model proposed in this paper still shows better performance and reliability.

Under the trend of economic globalization, crude oil futures plays the significant role in global economic growth. Therefore, predicting crude oil futures prices is extraordinarily essential for making better economic and investment decisions. First of all, the results of this paper can help investors and decision makers to make decisions fastly and carry out corresponding risk hedging operations, lowering the costs and losses related to the process of prediction. In addition, on account of the close relationship between crude oil futures and spot, Organization of the Petroleum Exporting Countries (OPEC) and International Energy Agency (IEA) can also apply the proposed method to formulate policies related to global crude oil price, which makes their decisions in the international energy market of great value. In particular, for some extreme events (including but not limited to the COVID-19 epidemic verified in this paper, the delivery price of crude oil futures fell to a negative value and the recent conflict between Russia and Ukraine pushed the crude oil futures

Table 8

Forecasting evaluation results of MDM test between WT-BLAC model and other models. (Training set: Test set = 4:6).

Models	MSE	MAPE	MAE	MDM_1	MDM_2	MDM_3
Panel A: WTI						
BiLSTM	$2.92 \times 10^{-22***}$	$2.17 \times 10^{-39***}$	$3.59 \times 10^{-37***}$	-10.4905	-15.1813	-14.5976
BLAC	0.0317	0.7434	0.8071	-2.1583	0.3277	-0.2444
CNN	$2.90 \times 10^{-35***}$	$2.13 \times 10^{-63***}$	$3.01 \times 10^{-55***}$	-14.0926	-21.4450	-19.3067
CNN-LSTM	$1.66 \times 10^{-11***}$	$2.79 \times 10^{-14***}$	$1.40 \times 10^{-13***}$	-6.9907	-7.9801	-7.7380
CNN-BiLSTM	$9.09 \times 10^{-10***}$	$1.15 \times 10^{-11***}$	$4.25 \times 10^{-11***}$	-6.3180	-7.0503	-6.8368
GRU	$3.48 \times 10^{-6***}$	$1.21 \times 10^{-5***}$	$3.59 \times 10^{-5***}$	-4.7252	-4.4481	-4.1929
LSTM	$4.38 \times 10^{-13***}$	$2.78 \times 10^{-16***}$	$3.22 \times 10^{-15***}$	-7.5640	-8.6475	-8.2973
LSTM-CNN	$8.31 \times 10^{-18***}$	$3.17 \times 10^{-48***}$	$1.03 \times 10^{-40***}$	-9.1363	-17.4820	-15.5274
BiLSTM-CNN	$7.40 \times 10^{-13***}$	$2.93 \times 10^{-33***}$	$1.99 \times 10^{-30***}$	-7.4831	-13.5574	-12.7924
BP	$5.03 \times 10^{-23***}$	$1.73 \times 10^{-38***}$	$1.34 \times 10^{-34***}$	-10.7138	-14.9441	-13.9157
SVR	$4.44 \times 10^{-20***}$	$8.80 \times 10^{-37***}$	$7.40 \times 10^{-33***}$	-9.8390	-14.4948	-13.4496
Panel B: Brent						
BiLSTM	$3.77 \times 10^{-12***}$	$1.68 \times 10^{-22***}$	$2.53 \times 10^{-22***}$	-7.2339	-10.5776	-10.5249
BLAC	0.0006	$1.19 \times 10^{-9***}$	$1.58 \times 10^{-8***}$	-3.4917	-6.2754	-5.8082
CNN	$1.72 \times 10^{-30***}$	$2.68 \times 10^{-54***}$	$3.14 \times 10^{-49***}$	-12.8380	-19.1406	-17.8101
CNN-LSTM	$9.11 \times 10^{-13***}$	$1.07 \times 10^{-24***}$	$9.30 \times 10^{-22***}$	-7.4571	-11.2156	-10.3577
CNN-BiLSTM	$7.13 \times 10^{-13***}$	$1.34 \times 10^{-24***}$	$7.09 \times 10^{-22***}$	-7.4953	-11.1874	-10.3926
GRU	$1.73 \times 10^{-10***}$	$1.73 \times 10^{-19***}$	$2.76 \times 10^{-16***}$	-6.6071	-9.6717	-8.6585
LSTM	$5.64 \times 10^{-23***}$	$1.07 \times 10^{-49***}$	$3.29 \times 10^{-42***}$	-10.7166	-17.9323	-15.9694
BiLSTM-CNN	$5.49 \times 10^{-27***}$	$1.76 \times 10^{-54***}$	$1.19 \times 10^{-47***}$	-11.8662	-19.1891	-17.3962
LSTM-CNN	$1.59 \times 10^{-24***}$	$4.27 \times 10^{-49***}$	$2.43 \times 10^{-43***}$	-11.1666	-17.7751	-16.2666
BP	$6.68 \times 10^{-24***}$	$5.95 \times 10^{-39***}$	$3.28 \times 10^{-35***}$	-10.9865	-15.1103	-14.1154
SVR	$3.88 \times 10^{-53***}$	$7.92 \times 10^{-133***}$	$5.19 \times 10^{-98***}$	-18.8355	-43.3215	-31.4459

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5%, 1% significance level respectively. MDM_1 , MDM_2 and MDM_3 represent the value of MDM test statistics under the loss function of MSE, MAPE and MAE respectively. H_0 : WT-BLAC model has the same predictive power as other models representing the null hypothesis. H_1 : WT-BLAC model has better predictive ability than the other model representing the alternative hypothesis.

Table 9

Forecasting evaluation results of MDM test between WT-BLAC model and other models. (Training set: Test set = 5:5).

Models	MSE	MAPE	MAE	MDM_1	MDM_2	MDM_3
Panel A: WTI						
BiLSTM	$1.14 \times 10^{-6***}$	$7.69 \times 10^{-13***}$	$1.47 \times 10^{-12***}$	-4.9837	-7.5433	-7.4398
BLAC	$6.60 \times 10^{-8***}$	$1.88 \times 10^{-14***}$	$1.57 \times 10^{-14***}$	-5.5625	-8.1200	-8.1477
CNN	$5.37 \times 10^{-21***}$	$1.00 \times 10^{-31***}$	$1.86 \times 10^{-29***}$	-10.2717	-13.4690	-12.8109
CNN-LSTM	$1.31 \times 10^{-13***}$	$3.11 \times 10^{-23***}$	$3.23 \times 10^{-21***}$	-7.8215	-10.9614	-10.3403
CNN-BiLSTM	$1.89 \times 10^{-13***}$	$4.98 \times 10^{-23***}$	$3.76 \times 10^{-21***}$	-7.7643	-10.8989	-10.3199
GRU	$6.37 \times 10^{-11***}$	$2.20 \times 10^{-16***}$	$8.86 \times 10^{-15***}$	-6.8196	-8.7819	-8.2341
LSTM	$1.97 \times 10^{-16***}$	$1.27 \times 10^{-26***}$	$4.17 \times 10^{-24***}$	-8.7982	-11.9793	-11.2263
LSTM-CNN	$9.99 \times 10^{-12***}$	$1.58 \times 10^{-27***}$	$4.89 \times 10^{-25***}$	-7.1288	-12.2459	-11.5063
BiLSTM-CNN	$6.79 \times 10^{-10***}$	$8.62 \times 10^{-20***}$	$5.72 \times 10^{-19***}$	-6.4104	-9.8923	-9.6301
BP	$7.80 \times 10^{-26***}$	$1.21 \times 10^{-46***}$	$9.31 \times 10^{-41***}$	-11.7446	-17.7279	-16.0506
SVR	$1.07 \times 10^{-20***}$	$6.51 \times 10^{-39***}$	$7.98 \times 10^{-35***}$	-10.1779	-15.5256	-14.3595
Panel B: Brent						
BiLSTM	$7.12 \times 10^{-15***}$	$1.67 \times 10^{-18***}$	$4.14 \times 10^{-19***}$	-8.2771	-9.4951	-9.6906
BLAC	$8.13 \times 10^{-6***}$	$2.18 \times 10^{-12***}$	$1.72 \times 10^{-11***}$	-4.5549	-7.3838	-7.0451
CNN	$1.03 \times 10^{-6***}$	$6.86 \times 10^{-5***}$	$2.26 \times 10^{-6***}$	-5.0073	-4.0477	-4.8393
CNN-LSTM	$2.33 \times 10^{-8***}$	$3.95 \times 10^{-9***}$	$9.96 \times 10^{-9***}$	-5.7667	-6.0984	-5.9272
CNN-BiLSTM	$1.20 \times 10^{-9***}$	$2.77 \times 10^{-11***}$	$6.99 \times 10^{-11***}$	-6.3138	-6.9656	-6.8094
GRU	$2.82 \times 10^{-5***}$	$5.02 \times 10^{-5***}$	0.0001***	-4.2627	-4.1249	-3.8585
LSTM	$2.08 \times 10^{-19***}$	$4.72 \times 10^{-30***}$	$1.22 \times 10^{-27***}$	-9.7862	-13.0193	-12.3095
BiLSTM-CNN	$4.66 \times 10^{-30***}$	$1.95 \times 10^{-58***}$	$2.41 \times 10^{-50***}$	-13.0211	-21.2713	-18.8811
LSTM-CNN	$3.46 \times 10^{-22***}$	$9.77 \times 10^{-42***}$	$4.39 \times 10^{-37***}$	-10.6612	-16.3928	-15.0559
BP	$3.21 \times 10^{-27***}$	$4.40 \times 10^{-45***}$	$1.16 \times 10^{-40***}$	-12.1846	-17.3561	-16.0835
SVR	$1.26 \times 10^{-42***}$	$5.71 \times 10^{-79***}$	$3.12 \times 10^{-66***}$	-16.6487	-27.7387	-23.6475

Note: *, ** and *** denote rejection of the null hypothesis at the 10%, 5%, 1% significance level respectively. MDM_1 , MDM_2 and MDM_3 represent the value of MDM test statistics under the loss function of MSE, MAPE and MAE respectively. H_0 : WT-BLAC model has the same predictive power as other models representing the null hypothesis. H_1 : WT-BLAC model has better predictive ability than the other model representing the alternative hypothesis.

soaring.), it also has better prediction performance than other models, which is very important for market participants to avoid risks in emergencies. Ultimately, apart from crude oil futures, the model proposed in this study can also be utilized to forecast other energy commodities, financial instruments or economic indicators.

CRediT authorship contribution statement

Yu Lin: Conception and design of study, Methodology, Analysis and interpretation of data, Revising the manuscript critically for important intellectual content, Supervision, Review editing.

Kechi Chen: Editing, Software visualization, Writing the original draft and interpretation of data, Drafting the manuscript. **Xi Zhang:** Acquisition of data, Review editing. **Bin Tan:** Conceptualization, Methodology, Investigation, Acquisition of data. **Qin Lu:** Conception and design of Study, Writing the original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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